SUPPLEMENTARY APPENDIX

AI-CoV Study: Autoimmune Encephalitis associated with COVID-19 and its vaccine –a Systematic review

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Definitions:

Adult: Age more than 18 years.

COVID-19 severity: Defined as per WHO clinical severity scale. [1]

Duration at presentation: The duration of symptoms prior to first medical contact.

Latency to neurological symptoms: The time to onset of at least one neurological symptom or sign from the index event (diagnosis of COVID-19 or its vaccination).

Latency to relapse or non-remission: Time to the onset of a new neurological symptom(s) or sign(s) or persistence of the presenting neurological symptom despite symptomatic therapy from the diagnosis of COVID-19, leading to suspicion of AIE and or initiation of immunotherapy.

Pediatrics: Age less than or equal to 18 years

Response to immunotherapy: Defined as clinical and or radiological improvement following initiation one of the standard immunotherapeutic agents which included steroids in the form of methylprednisolone (MP) or dexamethasone, intravenous immunoglobulin (IVIg), plasma exchange (PLEX), and monoclonal antibodies including rituximab.

Sequelae: Defined as the persistence of at least one of the neurological symptoms at the time of discharge or at follow-up

Treatment delay: Treatment delay was considered when the time to initiation of immunotherapy was more than the median latency to treatment initiation from the included studies.

Search strategy:

The search terms were strategized to bring into light the autoimmune encephalitis associated with COVID-19 as well as that related to COVID vaccine and the underlying pathophysiological mechanisms.

The first search comprised of Medical subject headings (MeSH) terms, "SARS CoV2" OR " COVID" OR "COVID 19" OR "COVID-19" were combined with "autoimmune encephalitis", OR, "limbic encephalitis" OR "autoimmune" OR "autoimmunity"OR "autoimmunity"OR "autoimmunities" OR "autoimmunization" OR "autoimmunizing" OR "encephalitis" OR "NMDA " OR "VGKC "OR "AMPA" OR "LGI 1" OR "CASPR" OR "gamma aminobutyric acid" OR "gamma aminobutyric" OR "gamma aminobutyric acid" OR "GABA" OR "encephalitis".

In the second search strategy, the MeSH terms, "SARS CoV2" OR " COVID" OR "COVID 19" OR "COVID-19" were combined with "encephalitis" to capture all the COVID encephalitis spectrum.

In the third search strategy, the MeSH terms, "SARS CoV2" OR " COVID" OR "COVID 19" OR "COVID-19" were combined with "vaccine" OR "vaccine" OR "vaccination" OR "vaccination" OR "vaccinable" OR "vaccinal" OR "vaccinate" OR "vaccinated" OR "vaccinates" OR "vaccinating" OR "vaccinations" OR "vaccination s" OR "vaccinator" OR "vaccinators" OR "vaccine s" OR "vaccined" OR "vaccines" OR "vaccines" OR "vaccine" OR "vaccins" OR "covid 19 vaccines" OR "covid 19" AND "vaccines" OR "covid 19 vaccines" OR "sarscov 2 vaccine" OR "covaxin" OR "BNT162b2" OR "mRNA-1273" OR "Ad26.COV2.S" OR "ChAdOx1".

In the fourth search strategy, the MeSH terms "pathogenesis" OR "mechanism" OR "pathophysiology" OR "process" OR "etiology" OR "pathogenesis" OR "mechanisms" OR "pathophysiologies" OR "physiopathology" OR "physiopathology" OR "processe" OR "processed" OR "processes" OR "processing" OR "processings" OR "aetiologie" OR "aetiologies" OR "aetiology" OR "etiologies" OR "etiology" OR "etiology" OR "causality" OR "causality"

Pathogenesis:

Further details of pathogenesis are provided here.[2,3]

Systemic infection, neurotropism and blood brain barrier dysfunction: The entry of SARS-CoV-2 into the CNS is mediated by a number of different pathways. The virus gains entry into the host cell via a complex interaction between its spike protein (S1) with angiotensin I converting enzyme 2 (ACE2) receptor and transmembrane serine protease 2 (TMPRSS2), in conjunction with basigin (BSG, i.e. EMMPRIN - extracellular matrix metalloproteinase inducer or CD147-cluster of differentiation 147 [4,5], Neuropilin-1 (NRP-1),[6,7] ADAM metallopeptidase domain 17 (ADAM17),[8,9] dipeptidyl peptidase 4 (DPP4) [10,11], angiotensin II receptor type 2 (AGTR2)[12], aminopeptidase N (ANPEP)[13], cathepsin,[14] and Furin.[15] This contributes to what is known as neurotropism, one of the key hallmarks of the virus. The other important pathways include retrograde axonal transport via olfactory mucosa, trans-synaptic transmission across the infected neurons, endothelial invasion and direct hematogenous spread to the CNS through the blood brain barrier deficient circumventricular organs.[16-19]

The interaction of spike protein of SARS-CoV-2 with its ACE receptors in the pneumocytes, lead to activation a plethora of inflammatory cascades, which ultimately contribute to cytokine storm.[20-23] The cytokine storm induced endothelial dysfunction leads to impairment of neurovascular unit (NVU), which subsequently contributes to blood brain barrier disruption.[17-21,24] Elevation of soluble endothelial markers (sE-selectin, tumor necrosis factor-α (TNF-α), interleukin 6, placental growth factor, and thrombomodulin and the hypoxia-inducible protein angiopoietin-like 4 (ANGPTL4)) in COVID-19 related encephalitis, as shown by Altmayer et al. provide direct evidence of endothelial dysfunction in these cases.[25] Several animal experiments have supported the notion of BBB disruption in COVID-19.

[26]Leaky vasculature in COVID-19 patients have also been found to be an important additional factor contributing to BBB disruption. Transport of intracellular virus through infected host immune cells (Trojan horse) represents another possible mechanism of neuro-invasion. [16-19,27-28]

Microglia activation, astrogliosis, neuronal damage and autoimmunity: Having entered the CNS, the virus infects the neighboring neurons, oligodendrocytes, astrocytes and microglia via ACE2, TMPRSS2 and other receptors.[29,30] The presence of S1 spike protein, triggers the activation of microglia, which lead to the release of inflammatory cytokines viz IL1β, TNF-α, IL-6, IL-10, IL-12, IL-17A, IL-23 and IL-33. In addition, there is consequent activation of ACE-2 receptor, HIF-1α , NOS, P2X7, NLRP-3and TNF-α expression. SARS-CoV-2 treated microglia are also known to cause enhanced Caspase 3/7 expression, apoptosis, mitochondrial activation, and reactive oxygen spices (ROS) production.[25,26] Yung et al. showed association of enhanced expression of complements like C1QC, CD74, FTL and FTH1, and downregulation of the homeostatic markers like P2RY12 with that of microglial activation.[31] Microgliosis and microglial nodules were found in 76% and 36%–44% of patients respectively, which have been thought to mediate mitochondrial dysfunction.[32] On the other hand, astrocytes have been found to function as viral 'replication hubs'.[33] In an autopsy series of 43 SARS-CoV2 infected patients, Matschke and collegues observed microglia activation and infiltration by cytotoxic T lymphocytes, which were most pronounced in the brainstem and cerebellum, and 79% of the patients had meningeal cytotoxic T lymphocyte infiltration.[34]

Neuronal damage and autoimmunity: Direct viral invasion mediated and glial cell activation mediated neuronal damage leads to exposure of immune privileged neuronal content to

activated immune system. In the lymph nodes these antibodies are presented to CD4+ T cells through dentritic cells, which activate naïve B cell. They subsequently differentiate into activate B cell and then to memory B cell and plasma cell. The memory B cells reach the brain, where they undergo restimulation, clonal expansion, maturation and differentiation into antibody-producing plasma cells. Differentiated plasma cells thereafter enters into brain parenchyma through leaky BBB. Thus, antibodies against the neuronal antigens are produced.[35-38] These autoantibodies are directed against cell surface and intracellular targets, i.e. NMDA receptor NMDAR), AMPA receptor (AMPAR), LGI-1, VKKC-CASPR2, Contactin 2, GABA A and B receptors and metabotropic Glutamate receptors (mGluR); the intracellular target antigens are Amphiphysin, Yo, Ri, Tr, Ma, CRMP5 etc. They play crucial role in transmission of neuronal signals and synaptic plasticity.[39]

Molecular mimicry & cross-reactive antibodies: The proteome of SARS-CoV2 comprises 29 proteins, consisting of 16 nonstructural (NSP), 4 structural proteins (SP), 9 open reading frames (ORF). [40] The structural similarities between the N-methyl-d-aspartic acid receptor (NMDAR) GluN1 and GluN2a subunits with the SARS-CoV-2 nonstructural protein 8 (NSP8) and 9 (NSP9), respectively, may induce immune-mediated cross-reactivity to the NMDAR. [39]

Genetic polymorphism: Recent studies have provided evidence of genetic polymorphisms contributing to severity of COVID-19. Some of which like HLA-C*04:01, HLA-DRB1*01:01, HLA-DRB1*04:01, HLA-A*11:01 are reported to be associated with severe SARS-CoV2 illness. On a similar backdrop, polymorphisms of HLA-DRB1*16:02 and HLA-DRB1*07 (DR7) have been shown to have association with anti-NMDA and anti-LGI1

encephalitis, respectively. These findings do warrant further studies to look for any such associations.[40-43]

Autoantibodies and cell surface receptor interaction:

Autoantibody- NMDAR complex leads to crosslinking of receptors, disruption of interaction between ephrin-B2 receptor and NMDAR. Subsequently, there is NMDAR internalization, hence synaptic availability of receptor density decreases.(127) Voltage gated potassium channels complex (VGKCs) comprising VGKC (Kv1), leucine-rich, glioma-inhibited 1 (LGI1), contactin-associated protein 2 (CASPR2), and contactin-2 are crucial in returning the cell to the resting state after an action potential is generated hence preventing hyperexcitation. LGI1 is a secreted synaptic protein that links between presynaptic ADAM23 (a disintegrin and metalloproteinase) and postsynaptic ADAM22 which is involved in transmission of inhibitory signals from the presynaptic potassium channel to the post synaptic α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic-acid (AMPA)-receptor. Autoantibodies against LGI1 decreases the LGI1-ADAM23/22 interaction and reversibly reduce postsynaptic AMPA-receptor density.[44] A cellular adhesion molecule (CAM) and part of the neurexin family, CASPR2 regulates the formation of axonal domains around the nodes of Ranvier which acts as a membrane scaffold that clusters Kv1 channels at the juxtaparanodal region. Autoantibodies to CASPR2 leads to limited clustering of Kv1 channels which leads to decrease in return of neuronal cell to baseline after an action potential.[45]

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